Jihoon Lee

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RESEARCH INTERESTS

I study wireless systems that preserve task performance under communication and computing constraints. My work spans semantic token communication, distributed vehicle control, and GPU sharing for AI-and-RAN systems.

Focus: Semantic token communication; RAN-aware GPU sharing; AI-and-RAN systems; V2X and distributed control; position-aided beam management.

EDUCATION

Ph.D. Program, Electrical & Electronic Engineering

Yonsei University · Mar 2021 - Present

RAMO Laboratory · Advisor: Prof. Seong-Lyun Kim

B.S., Electrical & Electronic Engineering

Yonsei University · Mar 2014 - Feb 2021

Seoul, South Korea

INTERNATIONAL PUBLICATIONS - PUBLISHED & ACCEPTED

[1] Semantics-Aware Hierarchical Token Communication: Clustering, Bit Mapping, and Power Allocation

Jihoon Lee, Seungeun Oh, Jihong Park, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Global Communications Conference (GLOBECOM), 2026. Accepted for presentation at IEEE GLOBECOM 2026.

[2] Enabling Distributed Control of Vehicle Platooning via Over-the-Air Consensus

Jihoon Lee, Yonghoon Jang, Hansol Kim, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Transactions on Wireless Communications, 2024. Vol. 23, No. 11, pp. 17205-17221.

https://doi.org/10.1109/TWC.2024.3451703

[3] Over-the-Air Consensus for Distributed Vehicle Platooning Control

Jihoon Lee, Yonghoon Jang, Hansol Kim, Seong-Lyun Kim, and Seung-Woo Ko

IEEE International Conference on Communications (ICC), 2023. Rome, Italy · pp. 5965-5971.

https://doi.org/10.1109/ICC45041.2023.10279484

[4] Stochastic Geometry Analysis of Position aided Beam Management

Jihoon Lee, Seung-Woo Ko, and Seong-Lyun Kim

ICTC Workshop on Post MIMO, 2023. Jeju, South Korea.

Conference paper

SUBMITTED MANUSCRIPTS

Hierarchical Token Communication: Semantics-Aware Representation, Protection, and Recovery

Jihoon Lee, Seungeun Oh, Jihong Park, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Transactions on Communications · **Submitted**

Dual-Stage Channel-Posterior Integration for MLM-Aided Token Communications

Hyelin Nam, **Jihoon Lee**, Seung-Woo Ko, Jinho Choi, and Seong-Lyun Kim

IEEE Wireless Communications Letters · **Submitted**

WORK IN PROGRESS

RAN-Aware Adaptive GPU Sharing for AI-and-RAN Systems

J. Lee, Y. Jo, Y. Lim, J. Kim, C. Lee, and S.-L. Kim

To be submitted to MobiCom 2027

RESEARCH PROJECTS

RAN-Aware Adaptive GPU Sharing for AI-and-RAN Systems

Ongoing · NVIDIA H100 testbed integrating OAI and NVIDIA Aerial/cuPHY for concurrent RAN processing and AI inference. The research compares MIG, MPS, and CUDA Green Contexts and uses RAN workload information to route AI requests between shared and disjoint GPU execution paths.

Development of a 5G-A vRAN Research Platform

Apr 2024 - Present · An integrated research platform that standardizes RU, DU, and CU interfaces and supports software-controlled, virtualized Open RAN systems.

Fundamental Technologies for Semantic Communication

Dec 2024 - Apr 2025 · Research on token-based SLM-LLM interaction to improve communication efficiency and inference accuracy.

Sub-THz Augmented Routing and Transmission for 6G

Dec 2021 - Dec 2023 · Research on routing and transmission foundations for next-generation sub-THz communication systems.

High-Accuracy Positioning-Enabled MIMO for 5G-V2X

Sep 2020 - Dec 2021 · A hardware/software testbed project for position-aided MIMO transmission and networking in 5G vehicle-to-everything services.

DOMESTIC CONFERENCES

English translations of Korean titles.

2025 · A Token-Based Approach to Highly Reliable Semantic Communication

Jihoon Lee et al. · KICS Winter Conference · Oral presentation

2023 · An Over-the-Air Consensus Algorithm for Distributed Vehicle Platooning Control

Jihoon Lee, Seung-Woo Ko, and Seong-Lyun Kim · JCCI · Oral presentation

2021 · Reinforcement-Learning-Based Resource Reallocation Control in Vehicular Networks

Jihoon Lee et al. · KICS Conference Proceedings · Conference paper

2021 · Position-Aware MAC Design for mmWave Vehicular Communications

Jihoon Lee, Seungeun Oh, Jinhyuk Choi, and Seong-Lyun Kim · JCCI · Oral presentation

2020 · Single-Channel Sharing in Massive IoT Using Zigbee

Jihoon Lee et al. · KIEES Summer Conference · Poster presentation

PATENTS

Selected Korean patents. Titles shown in English translation.

Apparatus and Method for Distributed-Consensus-Based Vehicle Platooning Control

KR 10-2023-0009373 · Application · Jihoon Lee and Seong-Lyun Kim

Method and Apparatus for Empty-Bottle Classification and Quantity Estimation Using Multi-Camera Sequential Machine Vision

KR 10-2613465 · Granted · Co-inventor

Method and Apparatus for Position-Information-Based Vehicular Communication

KR 10-2425933 · Granted · Seungeun Oh, Jihoon Lee, and Seong-Lyun Kim

Method and Apparatus for Resource Reallocation Control in Vehicular Networks

KR 10-2433577 · Granted · Seungeun Oh, Jihoon Lee, and Seong-Lyun Kim

TECHNICAL SKILLS & LANGUAGES

Python, MATLAB, System-level simulation, Stochastic geometry, Resource optimization, OAI / NVIDIA Aerial, CUDA Green Contexts

Korean: Native · English: Fluent